| | | | | |
|---|---|---|---|---|
| **3** | **(a)** | (numerically equal to) quantity of heat/(thermal) energy to change state/phase of unit mass | M1 | |
| | | at constant temperature | A1 | [2] |
| | | (*allow 1/2 for definition restricted to fusion or vaporisation*) | | |
| | **(b) (i)** | constant gradient/straight line *(allow linear/constant slope)* | B1 | [1] |
| | **(ii)** | $Pt = mL$ *or* power = gradient $\times L$ | C1 | |
| | | use of gradient of graph (*or two points separated by at least 3.5 minutes*) | M1 | |
| | | $110 \times 60 = L \times (372 - 325) \times 10^{-3} / 7.0$<br>$L = 9.80 \times 10^{5}\ \mathrm{J\,kg^{-1}}$ (*accept 2 s.f.*) (*allow 9.8 to 9.9 rounded to 2 s.f.*) | A1 | [3] |
| | **(iii)** | some energy/heat is lost to the surroundings *or* vapour condenses on sides | M1 | |
| | | so value is an overestimate | A1 | [2] |